

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5301

5 Credits: 4

Program Core

Source-grounded

Introduction to Machine Learning

□ To introduce the prominent methods for machine learning

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5301 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5301.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Robotic Process Automation
Course code	5301
Course title	Introduction to Machine Learning
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L: 4 T: 0 P: 0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

25 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- □ To introduce the prominent methods for machine learning
- □ To study the basics of supervised and unsupervised learning
- □ To study the basics of connectionist and other architectures

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding to interpret the concepts of supervised learning Compare the different dimensionality reduction	See official syllabus
CO2	14 Applying techniques Demonstrate the basic concepts of	See official syllabus
CO3	15 Applying classification	See official syllabus
CO4	Study neural network and markov processes 17 Applying	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2 3 3 2
CO2	CO2 3 2 3 3 3
CO3	CO3 3 2 3 3 2
CO4	CO4 3 3 3 3 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	supervised learning	6	As specified
Module 2	Compare the different dimensionality reduction techniques	5	As specified
Module 3	To Demonstrate the basic concepts of classification	7	As specified
Module 4	: Study neural network and markov processes	7	As specified

MODULE 1

supervised learning

This module is presented from the official syllabus scope for Introduction to Machine Learning. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: supervised learning
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding Machine

2 Understanding Machine is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Machine means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding Machine definition/meaning . steps, application . Technical terms English- .

▼ Learning applications - Learning associations

Learning applications - Learning associations is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Learning applications - Learning associations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, learning applications - learning associations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Learning applications - Learning associations means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-☐☐ Learning applications - Learning associations ☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐ ☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ Classification, Regression

Classification, Regression is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Classification, Regression using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classification, regression should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Classification, Regression means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- $\square\square$ Classification, Regression $\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ Unsupervised Learning

Unsupervised Learning is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Unsupervised Learning using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, unsupervised learning should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Unsupervised Learning means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഉപദേശിത ലേണിംഗ് (Supervised Learning) ന്റെ definition/meaning എന്താണ്. ഉപദേശിത ലേണിംഗ് ഉപയോഗിച്ച്, steps, application എങ്ങനെ പ്രയോജനപ്പെടുത്താം. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

▼ Reinforcement Learning. Supervised learning-

Reinforcement Learning. Supervised learning- is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Reinforcement Learning. Supervised learning- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, reinforcement learning. supervised learning- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Reinforcement Learning. Supervised learning- means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-റീനോർസ്മെന്റ് ലേണിംഗ്. Supervised learning- പര്യവേക്ഷിത ലേണിംഗ് definition/meaning പര്യവേക്ഷിത ലേണിംഗ്. പര്യവേക്ഷിത ലേണിംഗ് പര്യവേക്ഷിത, steps, application പര്യവേക്ഷിത പര്യവേക്ഷിത പര്യവേക്ഷിത. Technical terms English-റീനോർസ്മെന്റ് ലേണിംഗ് പര്യവേക്ഷിത ലേണിംഗ്.

▼ - Input representation, Hypothesis class 2 Understanding Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik- Chervonenkis (VC) Dimension

- Input representation, Hypothesis class 2 Understanding Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik- Chervonenkis (VC) Dimension is an official topic in Module 1 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify - Input representation, Hypothesis class 2 Understanding Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik- Chervonenkis (VC) Dimension using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, - input representation, hypothesis class 2 understanding introduction to machine learning, examples of machine learning applications - learning associations, classification, regression, unsupervised learning, reinforcement learning. supervised learning- input representation, hypothesis class, version space, vapnik- chervonenkis (vc) dimension should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What - Input representation, Hypothesis class 2 Understanding Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik- Chervonenkis (VC) Dimension means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ - Input representation, Hypothesis class 2 Understanding Introduction to Machine Learning, Examples of Machine Learning applications - Learning associations, Classification, Regression,

Unsupervised Learning, Reinforcement Learning. Supervised learning- Input representation, Hypothesis class, Version space, Vapnik- Chervonenkis (VC) Dimension ∞ ∞ definition/meaning ∞ ∞ . ∞ ∞ ∞ ∞ ∞ , steps, application ∞ ∞ ∞ ∞ . Technical terms English- ∞ ∞ ∞ ∞ ∞ .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Compare the different dimensionality reduction techniques

This module is presented from the official syllabus scope for Introduction to Machine Learning. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Compare the different dimensionality reduction techniques
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Probably Approximately Learning (PAC)

Probably Approximately Learning (PAC) is an official topic in Module 2 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Probably Approximately Learning (PAC) using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, probably approximately learning (pac) should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Probably Approximately Learning (PAC) means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- ϵ Probably Approximately Learning (PAC)
 ϵ definition/meaning ϵ . ϵ steps, application ϵ ϵ ϵ . Technical terms
 English- ϵ ϵ ϵ .

▼ Noise, Learning Multiple classes,

Noise, Learning Multiple classes, is an official topic in Module 2 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Noise, Learning Multiple classes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, noise, learning multiple classes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Noise, Learning Multiple classes, means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-നോയ്സ്, ലേണിംഗ് മൾട്ടിപ്പിൾ ക്ലാസ്സുകൾ, നിർവ്വചനം/അർത്ഥം, ഘട്ടങ്ങൾ, പ്രയോഗം. Technical terms English-ൽ ഉൾപ്പെടെ വിശദീകരിക്കുന്നു.

▼ Model Selection and Generalization

Model Selection and Generalization is an official topic in Module 2 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Model Selection and Generalization using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, model selection and generalization should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Model Selection and Generalization means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Model Selection and Generalization □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

▼ Dimensionality reduction- Subset selection

Dimensionality reduction- Subset selection is an official topic in Module 2 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Dimensionality reduction- Subset selection using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, dimensionality reduction- subset selection should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Dimensionality reduction- Subset selection means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Dimensionality reduction- Subset selection □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

▼ Principle Component Analysis 3 Applying Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis

Principle Component Analysis 3 Applying Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis is an official topic in Module 2 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Principle Component Analysis 3 Applying Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, principle component analysis 3 applying probably approximately learning (pac), noise, learning multiple classes, model selection and generalization, dimensionality reduction- subset selection, principle component analysis should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Principle Component Analysis 3 Applying Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Principle Component Analysis 3 Applying Probably Approximately Learning (PAC), Noise, Learning Multiple classes, Model Selection and Generalization, Dimensionality reduction- Subset selection, Principle Component Analysis എന്നെന്താണിത് എന്നതിന്റെ definition/meaning എന്താണിത്. എന്നാൽ എന്താണിത്, steps, application എന്നിവയെക്കുറിച്ച് എന്താണിത്. Technical terms English-ൽ എഴുതുക എന്നതാണ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

To Demonstrate the basic concepts of classification

This module is presented from the official syllabus scope for Introduction to Machine Learning. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: To Demonstrate the basic concepts of classification
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding methods

2 Understanding methods is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding methods means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Understanding methods
 definition/meaning . steps, application . Technical terms English-
 .

▼ K- fold cross validation

K- fold cross validation is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify K- fold cross validation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, k- fold cross validation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What K- fold cross validation means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- K- fold cross validation
 definition/meaning .
 steps, application . Technical terms English-
 .

▼ Boot strapping 2 Applying Measuring classifier performance- Precision,

Boot strapping 2 Applying Measuring classifier performance- Precision, is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Boot strapping 2 Applying Measuring classifier performance- Precision, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, boot strapping 2 applying measuring classifier performance- precision, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Boot strapping 2 Applying Measuring classifier performance- Precision, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Boot strapping 2 Applying Measuring classifier performance- Precision, definition/meaning. steps, application. Technical terms English-.

▼ 2 Applying recall,

2 Applying recall, is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying recall, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying recall, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying recall, means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 2 Applying recall, പദപരിച്ഛേദനം പദപരിച്ഛേദനം definition/meaning പദപരിച്ഛേദനം. പദപരിച്ഛേദനം പദപരിച്ഛേദനം, steps, application പദപരിച്ഛേദനം പദപരിച്ഛേദനം പദപരിച്ഛേദനം. Technical terms English- പദപരിച്ഛേദനം പദപരിച്ഛേദനം.

▼ ROC curves.

ROC curves. is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ROC curves. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, roc curves. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ROC curves. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ROC curves. ROC curve-
definition/meaning. ROC curve-
application. Technical terms English-
.

▼ Bayes Theorem, Bayesian classifier

Bayes Theorem, Bayesian classifier is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Bayes Theorem, Bayesian classifier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, bayes theorem, bayesian classifier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Bayes Theorem, Bayesian classifier means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Bayes Theorem, Bayesian classifier
 നിർവ്വചനം definition/meaning. ഘട്ടങ്ങൾ steps, application ഉദാഹരണം example. Technical terms English-ൽ നിർവ്വചിക്കുക.

▼ Maximum Likelihood estimation 3 Understanding Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression

Maximum Likelihood estimation 3 Understanding Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression is an official topic in Module 3 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Maximum Likelihood estimation 3 Understanding Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, maximum likelihood estimation 3 understanding classification- cross validation and re-sampling methods- k- fold cross validation, boot strapping, measuring classifier performance- precision, recall, roc curves. bayes theorem, bayesian classifier, maximum likelihood estimation regression should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Maximum Likelihood estimation 3 Understanding Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Maximum Likelihood estimation 3 Understanding Classification- Cross validation and re-sampling methods- K- fold cross validation, Boot strapping, Measuring classifier performance- Precision, recall, ROC curves. Bayes Theorem, Bayesian classifier, Maximum Likelihood estimation Regression ഏകദേശം definition/meaning എങ്ങനെ. ഏകദേശം ഏകദേശം ഏകദേശം, steps, application എങ്ങനെ എങ്ങനെ എങ്ങനെ. Technical terms English- എങ്ങനെ എങ്ങനെ.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

: Study neural network and markov processes

This module is presented from the official syllabus scope for Introduction to Machine Learning. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: : Study neural network and markov processes
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding Functions

2 Understanding Functions is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Functions means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-പാഠ്യ 2 Understanding Functions

പദപരിഭാഷണം പദപരിഭാഷണം definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-ൽ പദപരിഭാഷണം പദപരിഭാഷണം.

▼ Support Vector Machine

Support Vector Machine is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Support Vector Machine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, support vector machine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Support Vector Machine means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- $\square\square$ Support Vector Machine $\square\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ Discrete Markov Processes

Discrete Markov Processes is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Discrete Markov Processes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, discrete markov processes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Discrete Markov Processes means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Discrete Markov Processes

000000000000 00000 definition/meaning 000000000000. 000000 000000
 00000000, steps, application 000000 0000000000 000000. Technical terms
 English-0 00000 000000000000.

▼ Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation

Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, hidden markov models 2 understanding three basic problems of hmms- evaluation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐ Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation ☐☐☐☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐☐☐☐☐☐☐☐☐☐☐☐☐.

▼ 2 Understanding problem

2 Understanding problem is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding problem using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding problem should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding problem means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding problem $\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ Unsupervised Learning - Clustering Methods

Unsupervised Learning - Clustering Methods is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Unsupervised Learning - Clustering Methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, unsupervised learning - clustering methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Unsupervised Learning - Clustering Methods means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-☐☐ Unsupervised Learning - Clustering Methods ☐☐☐☐☐☐☐☐☐☐ definition/meaning ☐☐☐☐☐☐☐☐☐☐. ☐☐☐☐☐☐☐☐☐☐☐☐☐☐, steps, application ☐☐☐☐☐☐☐☐☐☐☐☐☐☐. Technical terms English-☐ ☐☐☐☐☐☐☐☐☐☐☐☐.

▼ **Expectation-Maximization Algorithm, 3 Understanding Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. References: 1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 2. Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004. 3. Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006 4. Mitchell. T, Machine Learning, McGraw Hill. 5. Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce**

Expectation-Maximization Algorithm, 3 Understanding Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. References: 1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 2. Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004. 3. Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006 4. Mitchell. T, Machine Learning, McGraw Hill. 5. Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce is an official topic in Module 4 of Introduction to Machine Learning. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Expectation-Maximization Algorithm, 3 Understanding Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. References: 1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 2. Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004. 3. Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006 4. Mitchell. T, Machine Learning, McGraw Hill. 5. Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, expectation-maximization algorithm, 3 understanding neural networks- the perceptron, activation functions. support vector machine . discrete markov processes, hidden markov models, three basic problems of hmms- evaluation problem, unsupervised learning - clustering methods - k-means, expectation-maximization algorithm. references: 1. christopher m. bishop, pattern recognition and machine learning, springer, 2006. 2. ethem alpaydın, introduction to machine learning (adaptive computation and machine learning), mit press, 2004. 3. margaret h. dunham. data mining: introductory and advanced topics, pearson, 2006 4. mitchell. t, machine learning, mcgraw hill. 5. ryszard s. michalski, jaime g. carbonell, and tom m. mitchell, machine learning : an artificial intelligence approach, tioga publishing company.nce should be

discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Expectation-Maximization Algorithm, 3 Understanding Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. References: 1. Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006. 2. Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004. 3. Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006 4. Mitchell. T, Machine Learning, McGraw Hill. 5. Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Expectation-Maximization Algorithm, 3 Understanding Neural Networks- The Perceptron, Activation Functions. Support Vector Machine . Discrete Markov Processes, Hidden Markov models, Three basic problems of HMMs- Evaluation problem, Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm. References: 1. Christopher M. Bishop, Pattern Recognition and Machine

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain 2 Understanding Machine. (3 marks)

► Define or explain Learning applications - Learning associations. (3 marks)

► Define or explain Classification, Regression. (3 marks)

► Define or explain Unsupervised Learning. (3 marks)

Module 2

► Define or explain Probably Approximately Learning (PAC). (3 marks)

► Define or explain Noise, Learning Multiple classes,. (3 marks)

► Define or explain Model Selection and Generalization. (3 marks)

► Define or explain Dimensionality reduction- Subset selection. (3 marks)

Module 3

► Define or explain 2 Understanding methods. (3 marks)

► **Define or explain K- fold cross validation. (3 marks)**

► **Define or explain Boot strapping 2 Applying Measuring classifier performance- Precision,. (3 marks)**

► **Define or explain 2 Applying recall,. (3 marks)**

Module 4

► **Define or explain 2 Understanding Functions. (3 marks)**

► **Define or explain Support Vector Machine. (3 marks)**

► **Define or explain Discrete Markov Processes. (3 marks)**

► **Define or explain Hidden Markov models 2 Understanding Three basic problems of HMMs- Evaluation. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Understanding Machine**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Probably Approximately Learning (PAC)**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **2 Understanding methods**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **2 Understanding Functions**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

- Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.
- Ethem Alpaydın, Introduction to Machine Learning (Adaptive Computation and Machine Learning), MIT Press, 2004.
- Margaret H. Dunham. Data Mining: introductory and Advanced Topics, Pearson, 2006
- Mitchell. T, Machine Learning, McGraw Hill.
- Ryszard S. Michalski, Jaime G. Carbonell, and Tom M. Mitchell, Machine Learning : An Artificial Intelligence Approach, Tioga Publishing Company.nce

Web resources listed in the official PDF

No separate web-resource heading was detected in the extracted text.

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTTR syllabus PDF for Course 5301. Verify institutional updates against the current official curriculum.